

# Shijian Xu

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## Education

- **Universität Basel** **Basel, Switzerland**  
Ph.D. student 04/2024 – 04/2028 (expected)  
Supervisor: Prof. Volker Roth
- **EPFL** **Lausanne, Switzerland**  
M.Sc. in Data Science 09/2020 – 02/2023  
Thesis: Weight Averaging for Out-of-Distribution Generalization and Few-Shot Domain Adaptation  
Supervisor: Prof. Grigorios Chrysos & Prof. Volkan Cevher
- **Nanjing University** **Nanjing, China**  
B.Sc. in Computer Science and Technology 09/2014 – 06/2018  
Thesis: Predicting Smartphone App Usage with Recurrent Neural Networks  
Supervisor: Prof. Wenzhong Li

## Publications & Preprints

- Temporal Self-Distillation: Faster Inference in Discrete Diffusion Language Models  
**Shijian Xu\***, Andrea Miele\*, Metod Jazbec, Volker Roth, Eric Nalisnick, Ilija Bogunovic  
Under review, \* equal contribution
- NIMO: a Nonlinear Interpretable MOdel  
**Shijian Xu**, Marcello Massimo Negri, Volker Roth  
ICLR 2026
- Simple Control Baselines for Evaluating Transfer Learning  
Andrei Atanov\*, **Shijian Xu\***, Onur Beker, Andrei Filatov, Amir Zamir  
arXiv, 2022
- Predicting Smartphone App Usage with Recurrent Neural Networks  
**S. Xu**, W. Li, X. Zhang, S. Gao, T. Zhan, Y. Zhao, W.-W. Zhu, T. Sun  
International Conference on Wireless Algorithms, Systems, and Applications (WASA), 2018

## Experience

- **University of Basel** **Basel, Switzerland**  
*Ph.D. research* 08/2024 – Current
  - Interpretable ML: a nonlinear yet interpretable model that preserves the transparency of linear models while capturing the complex representations enabled by neural networks, paper accepted in ICLR 2026
  - Accelerating diffusion LLMs (paper in submission)
- **Roche** **Basel, Switzerland**  
*PMDA Summer School 2024* 05/08/2024 – 09/08/2024
  - Predicting and understanding protein turnover in vitro and in vivo
- **LIONS Lab, EPFL** **Lausanne, Switzerland**  
*Master Thesis, supervisor: Dr. Grigorios Chrysos & Prof. Volkan Cevher* 09/2022 – 02/2023  
Out-of-Distribution Generalization
  - Improved weight averaging based method for OOD generalization and few-shot domain adaptation
- **DisneyResearch|Studios** **Zürich, Switzerland**  
*Research Intern, mentor: Dr. Christopher Schroers & Dr. Abdelaziz Djelouah* 02/2022 – 08/2022
  - Image super-resolution
- **VILAB, EPFL** **Lausanne, Switzerland**

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|-------------------------------------------------------------------------------------------|------------------------|
| Semester Project, supervisor: Prof. Amir Zamir                                            | 03/2021 – 12/2021      |
| Evaluation of Self-Supervised Learning                                                    |                        |
| - Proposed an evaluation standard for measuring the effectiveness of pre-training methods |                        |
| ○ <b>Tencent PCG</b>                                                                      | <b>Shanghai, China</b> |
| Research Intern                                                                           | 07/2021 – 09/2021      |
| - Transformer-based multi-label image classification                                      |                        |
| ○ <b>Tencent AI Lab</b>                                                                   | <b>Shenzhen, China</b> |
| Research Intern                                                                           | 02/2020 – 08/2020      |
| - Self-supervised multi-modal video classification                                        |                        |
| ○ <b>City University of Hong Kong</b>                                                     | <b>Hong Kong</b>       |
| Researcher                                                                                | 09/2018 – 12/2019      |
| - Deep learning for image specular highlight removal                                      |                        |
| - Semi-supervised single image heavy rain removal                                         |                        |
| ○ <b>Nanjing University</b>                                                               | <b>Nanjing, China</b>  |
| Project Student, supervisor: Prof. Wenzhong Li                                            | 09/2015 – 06/2018      |
| - LSTM based model for app usage prediction and recommendation                            |                        |

## Teaching

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Teaching assistant on the following courses:

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| ○ Scientific Computing (University of Basel)          | 2024, 2025 |
| ○ Seminar: Machine Intelligence (University of Basel) | 2025, 2026 |

## Awards

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|-----------------------------------------------------------------------|------|
| ○ Second Prize of Excellent Bachelor Thesis Award, Nanjing University | 2018 |
| ○ Third Prize of Elite Program Scholarship, Nanjing University        | 2017 |
| ○ Honorable Mention for MCM 2017, Nanjing University                  | 2017 |
| ○ Third Prize of People's Scholarship, Nanjing University             | 2016 |
| ○ Second Prize of People's Scholarship, Nanjing University            | 2015 |

## Languages

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- **English:** Advanced, TOEFL iBT 103 (2023)
- **German:** Beginner, B1
- **French:** Beginner, A1/A2
- **Chinese:** Native